

Educations

Ph.D. in Electrical and Electronic Engineering

Aug 2026 - present

Supervised by [Prof. Lihua Xie](#)

Nanyang Technological University ([NTU](#)), Singapore

Master Degree in Control Engineering and Automation

Jul 2024 - Jul 2026

Supervised by [Prof. Tung Lam Nguyen](#)

Hanoi University of Science and Technology ([HUST](#)), Hanoi, Vietnam

Thesis: Design control structures for parallel platforms in maritime applications

Bachelor Degree in Control Engineering and Automation

Oct 2020 - Apr 2024

Supervised by [Prof. Tung Lam Nguyen](#)

Hanoi University of Science and Technology ([HUST](#)), Hanoi, Vietnam

Thesis: Balancing, Motion Planning, and Tracking Control for Ballbot Systems

Work experiences

R&D Robotics Engineer at VinRobotics, Vingroup

Sep 2025 - Jul 2026

Full-time, supervised by [Dr. Minh Nhat Vu](#)

Hanoi, Vietnam

System identification, Kinematics calibration, Advanced control, impedance control, motion planning, optimization, obstacle avoidance, manipulator, humanoid, EtherCAT, ROS2, conduct scientific paper, etc.

Research Project Assistant at HUST

Oct 2021 - Dec 2026

Contract (Hybrid), supervised by [PhD. Chinh Hoang Duc](#) and [Assoc.Prof.PhD. Tung Lam Nguyen](#)

Hanoi, Vietnam

Safety control, Observer, Nonlinear control, Motion control/planning, ball-balancing robot, parallel mechanism, under-water vehicles, vibration compensation, etc.

Selected publications

- [8] **Vu, D. C.**, Nguyen, V. T., Nguyen, D. H, Nguyen, M. C., Tran, V. T., & Vu, M. N.(2026). Real-Time Conformal-Seeded Hybrid Inverse Kinematics for Offset Redundant Manipulators. *IEEE RA-L*. [\[pdf\]](#)
- [7] Nguyen, M. C.[†], **Vu, D. C.**[†], Vu, V. A., Pham, D. T., Nguyen, T., Phung, M. H., Ngo, A. V., & Vu, M. N. (2026). Adaptive Whole-Body MPC with Optimal Posture Planning for Humanoid Robots under Unknown Payloads. *IEEE-RAS 25th International Conference on Humanoid Robots (Humanoids 2026)*. [†]equal contribution
- [6] Nguyen, T. T. H., **Vu, D. C.**, Pham, M. D., Nguyen, T. L., & Nguyen, T. V. A. (2026). Unifying hierarchical sliding mode control and control barrier function for tilt angle constraint of a ball-balancing robot. *IET Cyber-Systems and Robotics*. [\[pdf\]](#)
- [5] **Vu, D. C.**[†], Tran, S.[†], Nguyen, T. L., & Hoang, D. C. (2025). Glocal trajectory generation and tracking control for AUVs with optimal coverage sensor networks. *Ocean Engineering*. [\[pdf\]](#) [†]equal contribution
- [4] **Vu, D. C.**, Nguyen, D. H., Vu, M. N., & Nguyen, T. L. (2025). Lagrangian-based modeling and safety-critical controls for Stewart platforms under marine operations. *Ocean Engineering*, 340, 122142. [\[pdf\]](#)
- [3] Pham, M. D., **Vu, D. C.**, Nguyen, T. T. H., Nguyen, T. V. A., Vu, M. N., & Nguyen, T. L. (2025). CBFs-based model predictive control for obstacle avoidance with tilt angle limitation for ball-balancing robots. *IEEE Access*. [\[pdf\]](#)
- [2] **Vu, D. C.**, Nguyen, D. H., & Nguyen, T. L. (2025). A novel approach of consensus-based finite-time distributed sliding mode control for Stewart platform manipulators motion tracking. *Results in Engineering*. [\[pdf\]](#)
- [1] **Vu, D. C.**, Pham, M. D., Nguyen, T. T. H., Nguyen, T. V. A., & Nguyen, T. L. (2024). Time-optimal trajectory generation and observer-based hierarchical sliding mode control for ballbots with system constraints. *International Journal of Robust and Nonlinear Control*. [\[pdf\]](#)

Honours & awards

NTU Research Scholarship

Nanyang Technological University (NTU), Singapore

Master, PhD Scholarship Programme

Vingroup Innovation Foundation (VINIF)

Best Thesis Defense Award

Hanoi University of Science and Technology

Academic activities

Invited review for *Nonlinear Dynamics, Ocean Engineering, IEEE Access, IEEE Robotics and Automation Letters (RA-L); International Conference on Intelligent Robots and Systems (IROS 2026), IEEE RO-MAN 2026, Humanoids 2026, International Conference on Robotics and Automation (ICRA 2027).*